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Portfolio

Martin Ngoh

Data Scientist | AI Engineer

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TECHNICAL SKILLS

Languages	Python, PySpark, SQL, R, SAS
ML & AI	PyTorch, TensorFlow, XGBoost, Prophet, Hugging Face, OpenAI/Claude APIs, LLM inference, guardrails
MLOps & Data	AWS (SageMaker, Textract), Docker, CDK, MLflow, FastAPI, PostgreSQL

EXPERIENCE

Senior Data Scientist <i>Accenture Federal Services</i>	July 2024 – Present <i>Arlington, VA</i>
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- Built and shipped a production LLM inference pipeline (AWS Textract with LLM prompting) that extracts structured JSON from unstructured documents and reconciles each record against a person database, an automated validation gate catching extraction errors before downstream systems; scaled throughput from 100 to 10,000+ documents/day and cut validation effort 100×.
- Designed, trained, evaluated, and deployed an XGBoost classification service reaching 88% accuracy on role mapping across heterogeneous sites, using n-gram features that generalize to unseen titles; deployed on SageMaker with CDK infrastructure as code, replacing 4-hour validation sessions with 30-second inference calls.
- Optimized a production Prophet forecasting workload for accuracy and cost simultaneously, reducing MAPE by 12% and cutting close to \$1M in annual compute spend.
- Mentored junior data scientists through code review and pairing while translating model behavior and failure modes for technical and non-technical stakeholders.

Data Scientist <i>Deloitte Consulting</i>	October 2021 – July 2024 <i>Arlington, VA</i>
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- Improved fraud-detection performance 8% using Bayesian optimization (BayesSearchCV) to tune an ensemble of neural-network, elastic-net, and gradient-boosting models.
- Instrumented centralized model tracking and experiment governance with MLflow across production ML initiatives, enabling reproducible evaluation and rapid iteration.
- Automated the model-training workflow in Python, cutting turnaround from 3 days to 2 hours (36× faster), and trained 20 colleagues on the new process.
- Analyzed customer-behavior data to identify AI/ML opportunities that informed a 3-year, \$30M product initiative.

PROJECTS

Justice Lens Police Accountability Analytics Python, Claude API, PostgreSQL, FastAPI, Docker	2026
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- Built and shipped a public analytics platform measuring MPD enforcement against demographic and reported-crime baselines, computing eight analyses live from DC Open Data across 509,000+ stops and 588,000+ incidents.
- Implemented veil-of-darkness testing and Bayesian search-threshold modeling to separate enforcement disparity from confounded stop-rate differences, publishing results whether they fault or favor MPD.
- Built a natural language map query on the Claude API resolving free-text questions into structured filters over the incident corpus; shipped on FastAPI, Docker, and Railway with 60+ tests.

SPEAKING & RECOGNITION

- Keynote panelist at Georgetown University's 2024 Black Alumni Summit. One of five AI experts addressing the implications of AI for the Black community before an audience of 200+.

EDUCATION

M.S. Business Analytics , Georgetown University (Varsity Men's Soccer Team)	2022
B.S. Supply Chain Management and Analytics , VCU (Varsity Men's Soccer Team)	2020